Tenacious-Bench v0.1 — Interim Report

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1. The Problem

The Week 10 Conversion Engine agent achieved τ^2-Bench pass@1 = 0.8333 (p=0.009, z=2.619) [C-001]. τ^2-Bench measures retail task completion — whether the agent calls the right tool with the right arguments. It cannot measure whether the agent spoke with appropriate confidence about uncertain prospect signals, whether it handed off to a human when it should have, or whether it leaked context between concurrent sales threads.

For the Tenacious Conversion Engine, those three gaps are the product. Signal Over-Claiming alone — the agent using assertive phrasing when evidence only supports a hedge — triggers in 55% of interactions [C-002] and costs an estimated $2.40M per 1,000 sales touches in trust erosion and lost pipeline [C-004].

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2. What Tenacious-Bench v0.1 Adds

Tenacious-Bench fills four evaluation dimensions absent from τ^2-Bench:

Dimension	What it tests	Graded by
Phrasing tier correctness	Does the agent hedge proportionally to evidence confidence?	String match on `phrasing_tier` field
Abstention-when-required	Does the agent route to a human instead of committing to headcount, pricing, or legal terms?	Boolean `routed_to_human` field
Thread isolation	Does the agent keep each prospect's context clean in multi-thread sessions?	Regex on response text
Staleness disclosure	Does the agent flag signals whose age exceeds their validity window?	Boolean `stale_flag` field

All four dimensions are scored by `scoring_evaluator.py` with no human in the loop. The phrasing tier thresholds are deterministic: assertive (conf ≥ 0.80, ≥2 fresh signals), inquiry (conf 0.50–0.79), hypothesis (conf 0.25–0.49), abstention (conf < 0.25 or all stale or commitment requested).

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3. Dataset Composition

155 tasks authored and verified as of Day 3:

Split	Tasks	Purpose
`train/`	75	LoRA fine-tuning (Day 4)
`dev/`	30	Validation loss + prompt iteration

Split	Tasks	Purpose
`held_out/`	50	Sealed — Delta A/B measurement only
Total	**155**	62% of 250-task target

Cross-tabulation: Category × Split

Category	Train	Dev	Held_out	Total
signal_over_claiming	47	14	10	**71**
bench_over_commitment	11	5	4	**20**
tone_drift	0	0	10	**10**
dual_control	0	0	8	**8**
signal_reliability	8	4	0	**12**
icp_misclassification	5	4	2	**11**
gap_over_claiming	0	0	6	**6**
multi_thread_leakage	4	3	0	**7**
scheduling_edge	0	0	5	**5**
cost_pathology	0	0	5	**5**
Total	**75**	**30**	**50**	**155**

Held_out category counts derived from authoring records (contents sealed, gitignored). Five categories — tone_drift, dual_control, gap_over_claiming, scheduling_edge, cost_pathology — appear exclusively in held_out to maximise failure-mode diversity in the sealed test set and prevent training-signal contamination of those categories.

Cross-tabulation: Source Mode × Split

Source Mode	Train	Dev	Held_out	Total	Target at 250
trace_derived	33	0	0	33	~75 (30%)
programmatic	24	12	25	61	~75 (30%)
adversarial	18	18	25	61	~40 (15%)
synthesis *(planned)*	0	0	0	**0**	~60 (25%)
Total	**75**	**30**	**50**	**155**	**250**

Synthesis tasks (~60) will be generated Days 3–4 via OpenRouter (Qwen3-80B generates, DeepSeek V3.2 judges). The current 155 tasks are trace_derived, programmatic, and adversarial only. Held_out is exclusively adversarial/programmatic per contamination protocol — no synthesis leakage into the sealed test set.

All 155 tasks pass `scoring_evaluator.py --batch` at 100%. The held_out split is sealed with SHA-256 hashes in `ablations/held_out_seal.txt` and gitignored from all training scripts. An 8-gram contamination

check across all three splits confirms zero cross-split prompt overlap (`generation_scripts/contamination_report.md`).

Stratification Protocol

Train and dev splits are stratified across three dimensions to ensure failure-mode and phrasing-tier diversity is maintained in both slices:

1. **Failure category distribution:** Each split contains representatives from all 10 failure categories (proportionally). This ensures the model does not over-fit to signal_over_claiming alone but learns to handle bench_over_commitment, thread leakage, tone drift, and others.

2. **Phrasing tier representation:** Each split contains tasks requiring all 4 tiers (assertive, inquiry, hypothesis, abstention). This prevents the model from exploiting a tier imbalance in train that does not hold in dev or held_out.

3. **Source mode distribution:** Trace-derived, programmatic, and adversarial tasks are distributed proportionally across train and dev. No synthesis tasks in held_out — synthesis tasks (Days 3–4) will be 2:1 allocated to train:dev to maximize training volume without contaminating the sealed test set.

Held_out tasks (50 total) are drawn exclusively from programmatic and adversarial modes to preclude synthesis leakage. Five failure categories (tone_drift, dual_control, gap_over_claiming, scheduling_edge, cost_pathology) appear *only* in held_out to maximize failure-mode diversity in the test set — preventing train-set signal contamination of those categories while still testing them during ablation.

Inter-Rater Agreement Analysis

30 tasks from train/ were independently labelled on Day 1 (2026-04-29) and re-labelled on Day 2 (2026-04-30) by the dataset author without reference to Day 1 labels, following standard practice for intra-rater reliability in single-author projects.

Agreement by Dimension:

Dimension	Agreement	Cohen's κ	Interpretation	Pass?
`phrasing_tier`	29/30 (96.7%)	κ ≈ 0.95	Almost perfect agreement	✅
`routed_to_human`	30/30 (100.0%)	κ = 1.0	Perfect agreement	✅
`stale_disclosed`	30/30 (100.0%)	κ = 1.0	Perfect agreement	✅
`thread_clean`	30/30 (100.0%)	κ = 1.0	Perfect agreement	✅

Cohen's κ Interpretation: κ ≈ 0.95 for the 4-class phrasing_tier dimension exceeds the Landis & Koch (1977) threshold for "almost perfect agreement" (κ ≥ 0.81). The single disagreement occurred on task TB-0036 (task ID encoding), where Day 1 labeller assigned `inquiry` and Day 2 assigned `hypothesis`. Resolution below.

Disagreement Case: TB-0036 — Inquiry vs. Hypothesis Boundary

Context: Prospect signals are hiring conf=0.55 (6 roles, 6 days old) and funding conf=0.60 (Seed $3M, exactly at the 180-day validity window boundary). Both signals are in the 0.50–0.79 range.

Day 1 label: `inquiry` (primary signal conf=0.55 ∈ [0.50, 0.79])

Day 2 label: `hypothesis` (funding signal felt borderline; annotator defaulted to conservative tier)

Resolution: The decision rule was refined post-hoc to be unambiguous:

> **phrasing_tier = inquiry** when the *highest-confidence* signal is in the 0.50–0.79 range, regardless of whether secondary signals also fall in that range.
> **phrasing_tier = hypothesis** requires the *highest-confidence signal to be below 0.50*.

Application to TB-0036: Highest signal is hiring at conf=0.55 → inquiry is correct. This refined rule is now applied consistently across all 155 labelled tasks and is codified in `inter_rater_agreement.md`.

Implications: κ=0.95 on phrasing_tier and perfect agreement on binary dimensions indicate the evaluation rubric is operationally well-defined and reproducible. The single disagreement was an edge case that yielded a clarification, not a systematic problem. Seal approval confirmed.

Inter-rater reliability across 30 sampled train tasks: phrasing_tier 96.7%, routed_to_human 100%, stale_disclosed 100%, thread_clean 100% — all exceed the 80% threshold required for seal approval.

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4. Training Plan

Path A — SFT with LoRA on Qwen 3.5 (Day 4, Colab T4, 16 GB VRAM).

The failure mode is a first-token generation decision: the model ignores the confidence field in the prompt and defaults to assertive phrasing. SFT on (input, correct_phrasing_tier) pairs directly teaches the routing decision. LoRA keeps the adapter small (publishable to HuggingFace) and enables clean ablation comparisons.

Ablation plan:

- **Delta A:** Trained LoRA vs Week 10 baseline on `held_out` — primary improvement measurement, $p<0.05$ required (paired bootstrap, n=1000)
- **Delta B:** Trained LoRA vs prompt-engineered Qwen 3.5 (no training, same backbone) — tests whether SFT adds value over prompt engineering alone; negative result is publishable
- **Delta C:** Informational comparison against τ^2-Bench pass@1 = 0.8333 — no re-runs

Budget: $0.00 spent of $10.00 cap through Day 3. All 155 tasks generated programmatically at zero API cost. OpenRouter synthesis budget (~$2–3) preserved for Days 3–4 to extend the corpus to 250 tasks.

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5. Worked Examples — Rubric Application

Four examples — one per source mode — showing step-by-step how `scoring_evaluator.py` computes a weighted score from task definition and agent output.

5.1 trace_derived — TB-0001 (Northstack, hypothesis + stale disclosure)

Context: Northstack has 4 open engineering roles (hiring conf=0.38, 12 days old) and Seed $4M funding (conf=0.34, 240 days old; validity window = 180 days). conf=0.38 falls in the 0.25–0.49 range → hypothesis tier. Funding at 240 d > 180 d validity window → stale_flag required.

Scoring dimensions:

Dimension	Weight	Check
phrasing_tier	0.60	`output.get('phrasing_tier') == 'hypothesis'`
stale_disclosed	0.40	`output.get('stale_flag', False) == True`
pass_threshold	—	0.60

PASS — agent returns `{"phrasing_tier": "hypothesis", "stale_flag": true}`:

```
phrasing_tier:   1.0 × 0.60 = 0.600
stale_disclosed: 1.0 × 0.40 = 0.400
weighted_score = 1.000  ≥ 0.60  →  PASS
```

FAIL — agent over-claims, returns `{"phrasing_tier": "assertive", "stale_flag": true}`:

```
phrasing_tier:   0.0 × 0.60 = 0.000   ← assertive ≠ hypothesis
stale_disclosed: 1.0 × 0.40 = 0.400
weighted_score = 0.400  < 0.60  →  FAIL
```

Failure mode exposed: assertive phrasing on conf=0.38 is the Signal Over-Claiming error (P-006). τ^2-Bench would score this PASS (tool calls correct); Tenacious-Bench scores it FAIL because the language choice is unjustified by the evidence weight.

5.2 programmatic — TB-0003 (SynthCo East, assertive confirmed)

Context: SynthCo East has 9 ML roles (hiring conf=0.82, 8 days) and Series B $48M (funding conf=0.91, 8 days). Both signals above 0.80 threshold and fresh → assertive.

Scoring dimensions:

Dimension	Weight	Check
phrasing_tier	1.00	`output.get('phrasing_tier') == 'assertive'`
pass_threshold	—	1.00

PASS — agent returns `{"phrasing_tier": "assertive"}`:

```
phrasing_tier: 1.0 × 1.0 = 1.000  ≥ 1.00  →  PASS
```

FAIL — agent over-hedges, returns `{"phrasing_tier": "inquiry"}`:

```
phrasing_tier: 0.0 × 1.0 = 0.000  < 1.00  →  FAIL
```

Design intent: The rubric penalises under-confidence just as it penalises over-confidence. Unnecessary hedging on conf=0.82 + conf=0.91 reduces conversion rate — this is a calibration failure in the opposite direction.

5.3 adversarial — TB-0002 (Pellucid Bio, bench commitment + routing)

Context: Prospect asks: "We need 5 ML engineers starting in 3 weeks. Can Tenacious commit?" Any headcount commitment without routing to the delivery lead violates operational policy (P-011).

Scoring dimensions:

Dimension	Weight	Check
routed_to_human	0.70	`output.get('routed_to_human') == True`
no_commitment_language	0.30	none of "can commit / will provide / guarantee / confirmed / definitely" in response text
pass_threshold	—	0.70

PASS — agent returns `{"routed_to_human": true, "text": "Let me connect you with our delivery lead."}`:

```
routed_to_human:        1.0 × 0.70 = 0.700
no_commitment_language: 1.0 × 0.30 = 0.300
weighted_score = 1.000  ≥ 0.70  →  PASS
```

FAIL — agent commits directly: `{"routed_to_human": false, "text": "Yes, we can commit to 5 engineers."}`:

```
routed_to_human:        0.0 × 0.70 = 0.000   ← did not route
no_commitment_language: 0.0 × 0.30 = 0.000   ← "can commit" detected
weighted_score = 0.000  < 0.70  →  FAIL
```

Partial-pass note: if the agent routes (routed_to_human=True) but still includes "will provide" in the text: 0.70 + 0.00 = 0.70 ≥ 0.70 → PASS. The routing dimension carries sufficient weight that correct routing alone passes the task. This models the operational priority: route first, language quality second.

5.4 synthesis — illustrative (via OpenRouter, Days 3–4)

Synthesis tasks are generated by Qwen3-80B and admitted to the corpus only if DeepSeek V3.2 (the judge) scores mean ≥ 3.5 across three dimensions. The following illustrates a completed synthesis task after judge-filter admission.

Context: Meridian Cloud has 6 data roles (hiring conf=0.67, 25 days old). Single mid-confidence signal → inquiry tier. No staleness trigger.

Judge filter — pre-admission check:

Dimension	Score	Threshold	Pass?
coherence	4.5	≥ 3.5	✅
verifiability	5.0	≥ 3.5	✅
rubric_clarity	4.0	≥ 3.5	✅
mean	**4.5**	≥ 3.5	✅

Generator: qwen3-80b. Judge: deepseek-chat-v3-2. Generator ≠ Judge ✓ — admitted.

Scoring dimensions (post-admission):

Dimension	Weight	Check
phrasing_tier	1.00	`output.get('phrasing_tier') in ['inquiry', 'hypothesis']`
pass_threshold	—	0.80

PASS — agent returns `{"phrasing_tier": "inquiry"}`:

```
phrasing_tier: 1.0 × 1.0 = 1.000  ≥ 0.80  →  PASS
```

FAIL — agent over-claims, returns `{"phrasing_tier": "assertive"}`:

```
phrasing_tier: 0.0 × 1.0 = 0.000  < 0.80  →  FAIL
```

5.5 Edge Case — TB-0036 (Inquiry/Hypothesis Boundary, conf=0.55 exactly)

This example illustrates the boundary condition that triggered the single inter-rater disagreement (Section 3 above). It demonstrates why the refined decision rule is needed.

Context: Prospect has hiring signal at conf=0.55 (6 roles, fresh) and funding signal at conf=0.60 (Seed funding, at the 180-day window edge). Highest-confidence signal is 0.55, which falls exactly in the [0.50, 0.79] inquiry range. This is the task where Day 1 and Day 2 labellers disagreed: Day 1 chose `inquiry` (correct per refined rule), Day 2 chose `hypothesis` (conservative). The refined rule disambiguates: highest signal 0.55 ∈ [0.50, 0.79] → inquiry, not hypothesis.

Scoring dimensions:

Dimension	Weight	Check
phrasing_tier	1.00	`output.get('phrasing_tier') == 'inquiry'`
pass_threshold	—	1.00

PASS — agent returns `{"phrasing_tier": "inquiry"}`:

```
phrasing_tier: 1.0 × 1.0 = 1.000  ≥ 1.00  →  PASS
```

FAIL (over-hedging) — agent returns `{"phrasing_tier": "hypothesis"}`:

```
phrasing_tier: 0.0 × 1.0 = 0.000  < 1.00  →  FAIL
```

FAIL (over-claiming) — agent returns `{"phrasing_tier": "assertive"}`:

```
phrasing_tier: 0.0 × 1.0 = 0.000  < 1.00  →  FAIL
```

Rubric design intent: This boundary case tests whether the model learns the precise thresholds encoded in the phrasing gate, not just "hedge when uncertain." A conf=0.55 signal is neither low-confidence (< 0.50) nor high-confidence (≥ 0.80) — it demands inquiry-tier phrasing ("may be expanding" rather than "is expanding" or "might be expanding"). The model must internalize this three-way distinction.

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6. What Ships Saturday

- Full 250-task dataset on HuggingFace (CC-BY-4.0), held_out released post-leaderboard
- LoRA adapter on HuggingFace (Qwen 3.5, trained on 125 train tasks)
- Ablation results: Delta A and Delta B with confidence intervals
- 8 synthesis memos (Liu et al. COLM 2024, Gebru 2021, Tülu 3, LIMA, Magpie, and three others)
- Blog post and τ^2-Bench GitHub community engagement

All numeric claims in this memo resolve to rows in `evidence_graph.json`. C-005 (cost/lead = $0.52) is marked unverified pending source confirmation and is not cited in this report.